VU BAO KHANH

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PROFESSIONAL SUMMARY

Fullstack Developer with hands-on experience building web applications using React, TypeScript, NestJS, and PostgreSQL through personal projects involving real-time systems, asynchronous processing, authentication, and payment integration. Familiar with Docker, Redis, Socket.IO, and CI/CD workflows. Currently developing and maintaining client websites in a professional PHP environment, with experience across development, deployment, and production troubleshooting.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP

Frontend: React, Next.js, Tailwind CSS, Mantine UI, Zustand, TanStack Query, dnd-kit

Backend: Node.js, NestJS, Prisma ORM, RESTful APIs, Socket.IO, BullMQ

Databases & Caching: PostgreSQL, MySQL, Redis

Tools & Technologies: Docker, Git, JWT, OAuth 2.0, FCM, FFmpeg, HLS.js, Cloudflare R2

PROFESSIONAL EXPERIENCE

PHP Developer - BM WEB Co., Ltd | *Mar 2026 - Present*

- Developed and customized numerous client websites using PHP, MySQL, JavaScript, and AJAX, translating business requirements from the Business team into technical implementation.
- Optimized website performance and SEO — improving slow MySQL queries, implementing caching strategies, reducing page load time, and applying on-page SEO best practices (meta tags, sitemap, structured data).
- Ensured responsive, cross-browser compatible interfaces, delivering consistent user experience across devices and browsers.
- Managed the full website lifecycle from development and deployment to ongoing maintenance, feature updates, and bug fixes, including SMTP email configuration for contact forms and notifications.
- Deployed and maintained production environments: configured domains, DNS, SSL certificates, databases, and cPanel hosting; diagnosed and resolved issues across both application and server layers to ensure site stability.

Frontend Developer Intern - Cantho University Software Center | *May 2025 - Aug 2025*

- Developed responsive web interfaces using Next.js, Mantine UI, and Tailwind CSS.
- Integrated RESTful APIs with Axios to display and update application data.
- Built reusable UI components and implemented frontend state handling.
- Collaborated with backend developers to integrate APIs, debug issues, and deliver assigned features.

SELECTED PROJECTS

StreamHub - Video Streaming Platform | Fullstack Developer | *May 2026 - Present*

Stack: React, TypeScript, NestJS, PostgreSQL, Prisma ORM, Redis, BullMQ, FFmpeg, HLS.js, Cloudflare R2, Zustand, TanStack Query, Docker

- Developed a full-stack video streaming platform using React, NestJS, PostgreSQL, and Prisma, including user-facing features and an admin management system.
- Designed a decoupled video processing architecture, separating CPU-intensive transcoding tasks from the API server using BullMQ, Redis, and a standalone worker.
- Built an asynchronous video transcoding pipeline using BullMQ and FFmpeg to convert uploaded videos into Adaptive Bitrate HLS (360p–1080p) and store streaming assets on Cloudflare R2.
- Developed a custom HLS.js video player with adaptive quality selection, episode navigation, and cross-device watch-progress synchronization.
- Implemented JWT Refresh Token Rotation, Google OAuth 2.0, multi-device session management, and dynamic RBAC for authentication and permission-based access control.
- Integrated PayOS (VietQR) for subscription payments with signature verification and idempotent webhook handling to prevent duplicate transaction processing.
- Optimized frontend state and data fetching with Zustand and React Query, and containerized the API, transcoder, PostgreSQL, and Redis services using Docker.

Demo: https://streamhub.tforge.life

Krello - Real-Time Collaborative Task Management | Fullstack Developer | *Sep 2025 - Dec 2025*
Stack: React, TypeScript, NestJS, PostgreSQL, Prisma ORM, Redis, Socket.IO, Zustand, @dnd-kit, JWT, FCM, Docker

- Developed a Trello-inspired fullstack task management platform, building RESTful APIs with NestJS/Prisma and interactive Kanban interfaces with React and @dnd-kit.
- Built real-time collaboration using Socket.IO and Redis, synchronizing board/card updates, comments, typing indicators, and multi-connection user presence across clients.
- Implemented drag-and-drop card/list ordering with position-based reordering while minimizing unnecessary database updates.
- Designed authentication and authorization with JWT access/refresh token rotation, device-based session management, Google OAuth 2.0, board-level permissions, and RBAC.
- Built a concurrency-safe Axios silent-refresh queue to handle simultaneous 401 responses and automatically retry pending requests after token renewal.
- Optimized data access with Prisma query optimization and Redis caching; integrated FCM push notifications, transactional emails, and image processing with Multer/Sharp.
- Automated deployment to an Ubuntu VM (Proxmox) using GitHub Actions, a self-hosted runner, SSH, PM2, and Cloudflare DNS.

Demo: https://krello.click/
GitHub: https://github.com/vubaokhanh2311

EDUCATION

FPT Polytechnic Can Tho | Web Development | Aug 2023 - Aug 2025 | GPA: 3.75/4.0